



Technical Service Bulletin

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Procedure 002-004, Cylinder Head

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This Service/Parts Topic revises Procedure 002-004, Cylinder Head to include the intake valve recession limits.

In the past few years, there has been an issue with conflicting information in regard to ISM/QSM intake valve recession limits.

As field locations have worked their way through valve beat-in issues, it has become apparent that there is some conflicting information within Service Parts Topics, PITS documents and old Thursday Notes.

Valve beat-in and associated progressive damage such as cam bushing failures, cam lobe failures, cam follower damage, and valve failures can be significantly influenced by valve seals, lube oils, and maintenance practices.

During the years 1998 and 1999, a group of automotive engines were built with dry valve seals. The Engine Serial Numbers (ESNs) and dates are given below:

- Beginning with ESN 34891154 (December 7, 1997)
- Ending with ESN 34929771 (September 20, 1998)
- Engines built with the dry seal can experience intake seat wear (beat-in) due to lack of lubrication during extended periods of high boost operation. This beat-in leads to negative intake lash that causes the valve train to be continuously loaded with an additional load spike during combustion. The contact stresses on the opening ramp lead to asperity scale spalling (frosting) and camshaft lobe failure.

Several improvements have been made since that time period.

- Improved valve seal (Part Number 4003966) went into production on September 20, 1998.
- J120V valve seat inserts were implemented in production in the intake positions with ESN 35062901 (June 24, 2001). J120V valve seat inserts had already been used in the exhaust locations.

Related Procedures

35-002-004 (/qs3/pubsys2/xml/en/procedures/35/35-002-004.html)	Cylinder Head (/qs3/pubsys2/xml/en/procedures/35/35-002-004.html)
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